

Enterprise Data Simulation – Design Round

Business Analysis Track

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Role Applied: Business Analyst + Data Engineer Assessment

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Introduction

As a Business Analyst, my primary responsibility is to understand business requirements, define meaningful business metrics, and ensure that data is transformed into actionable insights for decision-makers. In this case, the objective is to evaluate how trade promotions impact weekly sales and inventory across different regions. My role is to define promotion effectiveness, identify the required business data, recommend the appropriate level of analysis, and ensure all stakeholders have a common understanding of the metric before it is used for reporting.

A1. Metric Definition – Weekly Promotion Effectiveness

Definition

Weekly Promotion Effectiveness measures the success of a trade promotion by comparing sales performance during promotional weeks with normal (baseline) sales while also considering changes in inventory levels.

This metric helps determine whether a promotion generated additional sales without creating excess inventory or stock shortages.

Formula

Promotion Effectiveness (%)

$$= ((\text{Sales During Promotion} - \text{Baseline Sales}) \div \text{Baseline Sales}) \times 100$$

Where:

- **Sales During Promotion** = Total sales during the promotional week.
- **Baseline Sales** = Average weekly sales during previous non-promotional weeks for the same SKU or category in the same region.

Supporting KPIs

- Weekly Sales Growth (%)
- Inventory Reduction (%)
- Sell-through Rate
- Promotion Participation Rate
- Promotion ROI (if promotional cost is available)

Business Value

This metric enables leadership to:

- Measure whether promotions increased sales.
- Identify products that respond well to promotions.
- Detect overstocking or stock-out situations.
- Compare promotion performance across products and regions.
- Improve planning for future promotional campaigns.

A2. Analytical Questions

Leadership may want answers to the following business questions:

1. Which promotions generated the highest increase in weekly sales?

Purpose: Identify the most successful campaigns and replicate them in future promotions.

2. Which product categories respond best to promotions?

Purpose: Understand customer buying behavior and optimize promotional investments.

3. Which regions experienced the largest inventory reduction during promotional weeks?

Purpose: Measure whether promotions successfully cleared inventory.

4. Are there regions where inventory remained high despite promotions?

Purpose: Identify ineffective campaigns or demand forecasting issues.

5. How do SKU-level promotions compare with Category-level promotions?

Purpose: Determine which promotion strategy delivers better business results.

A3. Required Data Elements

POS Sales

- Date
- Product Key
- Region Key
- Promotion Key
- Quantity
- Sales Amount

Inventory Data

- Date
- Product Key
- Region Key
- Inventory Quantity

Promotion Master

- Promotion Key
- Promotion Name
- Start Date
- End Date
- Level (SKU/Category)
- Discount Percentage

Product Master

- Product Key
- SKU
- Product Name
- Category
- Subcategory
- Brand

Region Master

- Region Key
- Region
- State

A4. Granularity

Promotion effectiveness should be measured at the following level:

Week × Region × SKU × Promotion

Reasoning

This level of granularity provides sufficient detail to evaluate promotional performance while avoiding unnecessary complexity.

Benefits include:

- Accurate weekly sales comparison.
- Region-specific performance analysis.
- SKU-level promotion tracking.
- Easy aggregation to category, brand, or national level.
- Better support for management dashboards.

Using a finer grain (daily) may introduce unnecessary noise, while using a higher grain (monthly) could hide important weekly trends and promotional impacts.

A5. Stakeholders and Business Decisions

Stakeholder	Business Decision
Sales Team	Identify promotions that increase sales and improve future sales strategies.
Marketing Team	Plan future promotional campaigns and optimize promotional budgets.
Supply Chain Team	Ensure adequate inventory during promotions and reduce stock shortages.
Inventory Manager	Prevent overstocking and improve inventory turnover.
Senior Leadership	Evaluate overall business performance and allocate investment to the most effective promotions.

Stakeholder Perspective

1. Marketing Team

Definition of Success

- Maximum increase in sales.
- Higher customer participation.
- Better promotion reach.

Possible Assumptions

- Every increase in sales is caused by promotions.
- All regions respond similarly.

2. Supply Chain Team

Definition of Success

- Stable inventory levels.
- No stock-outs.
- Reduced excess inventory.

Possible Assumptions

- Sales increases should not create inventory shortages.
- Inventory should remain balanced throughout the promotion.

Validation with Stakeholders

Before implementing this metric in dashboards, I would:

- Conduct workshops with business stakeholders.
- Review historical promotional reports.
- Validate business definitions with Sales, Marketing, and Supply Chain teams.
- Agree on the baseline sales calculation method.
- Define rules for overlapping promotions.
- Obtain final stakeholder approval before implementation.

Ensuring Alignment Across Teams

To maintain consistency:

- Create a Business Metric Definition Document.
- Standardize KPI calculations.
- Maintain a centralized business glossary.
- Establish data quality validation rules.
- Schedule periodic stakeholder reviews for metric validation.

Section C – Design Trade-offs

1. Simplicity over Flexibility

I selected a weekly reporting frequency instead of daily reporting.

Reason: Weekly reporting aligns with distributor sales data, simplifies analysis, reduces reporting complexity, and is easier for leadership to interpret.

Trade-off: Daily insights are sacrificed in favor of a simpler and more consistent analytical model.

2. Performance over Analytical Accuracy

I recommend using pre-calculated baseline sales instead of recalculating them every time a dashboard is refreshed.

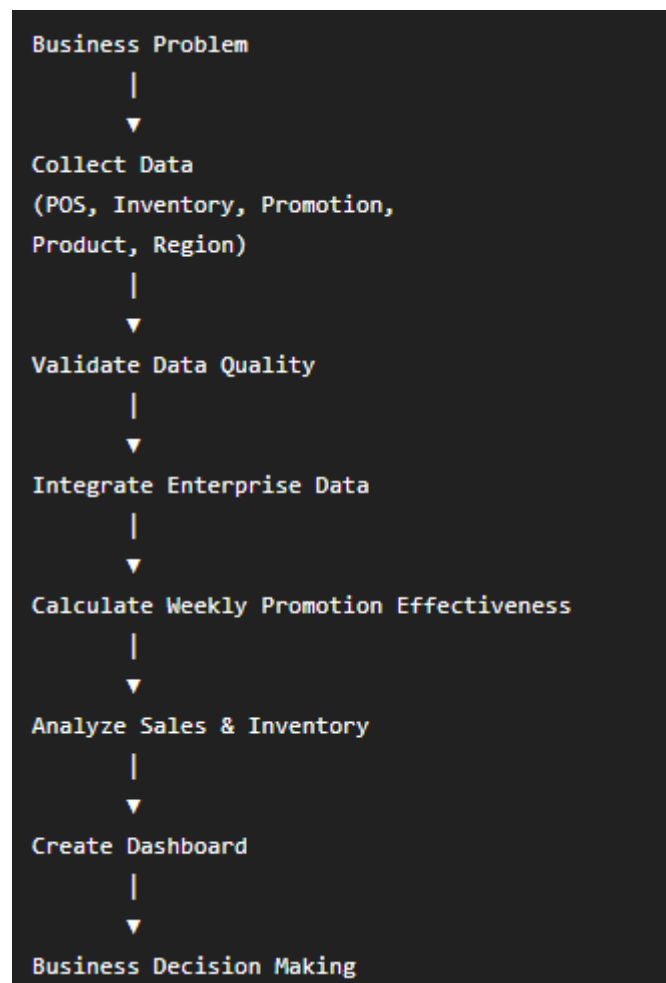
Reason: This significantly improves dashboard performance and reduces report loading time.

Trade-off: Small historical corrections may not immediately appear until the baseline is refreshed.

Conclusion

The proposed design provides a standardized approach for evaluating trade promotion effectiveness by integrating sales, inventory, product, promotion, and regional information into a single business metric. Measuring promotion effectiveness at the **Week × Region × SKU × Promotion** level enables leadership to compare campaign performance, identify successful promotional strategies, optimize inventory planning, and make data-driven decisions. Through clear metric definitions, stakeholder alignment, and standardized business rules, this solution ensures consistent reporting and supports effective business planning across departments.

Business Process Flow



Stakeholder Mapping

